

EXPANDING SPACE

22–23 The scale of the universe

34–37 Radiation

The Big Bang 48–51

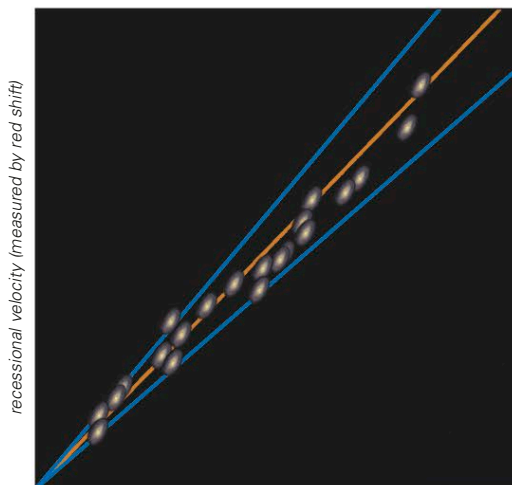
The fate of the universe 58–59

Galaxy clusters 326–327

A CRUCIAL PROPERTY of the universe is that it is expanding. It must be growing, because distant galaxies are quickly receding from Earth and more distant ones are receding even faster. Assuming that the universe has always been expanding, it must once have been smaller and denser—a fact that strongly supports the Big Bang theory of its origin.

MEASURING EXPANSION

For the relatively nearby universe, the expansion can be calculated by comparing the distances to remote galaxies and the speeds at which they are receding. The galaxies' velocities are measured by examining the red shifts in their light spectra (see p.35). Their distances are calculated by detecting a class of stars called Cepheid variables in the galaxies and measuring the stars' cycles of magnitude variation (see pp.282, 313). For more remote parts of the universe, the expansion rate is measured by methods such as analysis of tiny fluctuations in the cosmic microwave background radiation (see pp.51, 54). The result is a number known as the Hubble constant—an expression of the universe's expansion rate. The value of the constant is about 50,000 mph (81,000 kph) per million light-years for the relatively nearby universe and a little less for the most remote observable objects. This means, for example, that two relatively nearby galaxies situated 1 billion light-years apart will be receding from each other at 50 million mph (81 million kph). On a familiar time scale, this is actually a very gradual expansion—an increase in the galaxies' distance of 1 percent takes tens of millions of years.



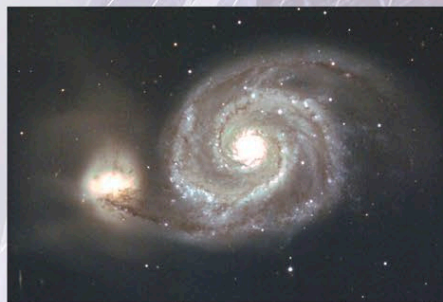
distance from Earth (measured by variable stars)

HUBBLE CONSTANT

The recession velocity of remote galaxies rises with distance, and this relationship forms a straight line on a graph. Estimates of the line's slope yield values of the Hubble constant.

THE NATURE OF EXPANSION

Several notable features have been established about the universe's expansion. First, although all distant galaxies are moving away, neither Earth nor any other point in space is at the center of the universe. Rather, everything is receding from everything else, and there is no center. Second, at a local scale, gravity dominates over cosmological expansion and holds matter together. The scale at which this happens is surprisingly large—even entire clusters of galaxies resist expansion and hold together. Third, it is incorrect to think of galaxies and galaxy clusters moving away from each other “through” space. A more accurate picture is that of space itself expanding and carrying objects with it. Finally, the expansion rate almost certainly varies. Cosmologists are greatly interested in establishing how the expansion rate may change in the future. The future rate of expansion will decide the eventual fate of the universe (see pp.58–59).



LOCAL GRAVITY

The galaxies above are not moving apart. They will continue to collide despite cosmological expansion. Galaxy clusters are also held together by gravity.

